PEER → VECTOR: Status Update — Tenure (Setting 3)

Date: 2026-06-03
Companion doc: `Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md` (Setting 2 / basketball)
Purpose: Give **VECTOR** a ground-truth snapshot of where the **academia / tenure** empirical setting stands so Charles can decide how to represent PEER in the manuscript and in any briefings.
Author: PEER (Cursor agent on Cursor Workspace PDE)

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1. Executive Summary

Topic	Status
Pipeline (scrape → parse → panel → pool metrics)	**Complete end-to-end** for the current corpus
Inverted-U signal	**Yes — present in preliminary analysis** (stage 9; see §4)
Cox survival model	**Wired and runnable** (Cell 10); z-scored covariates in Cell 10.5
Coverage completeness	**Partial** — pipeline covers ~60+ R1 CS departments with usable data; full 187-school roster is a publication-time goal
OpenAlex linkage	**API-based** (not bulk); ~30–40% HIGH confidence matches; MULTI and NONE coverage is a known limitation
Formal model results ready to quote	**Not yet** — preliminary binned curve is the current artifact; regression/Cox output is the next step

Bottom line for VECTOR: The tenure setting has a **working pipeline, a longitudinal panel, and a preliminary empirical result consistent with the inverted-U hypothesis**. It is not as polished as Army/CODA or basketball/SCOUT. The right framing for the manuscript is “preliminary third-setting replication” — honest about coverage limitations, with the stage 9 figure as the anchor.

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2. What Setting 3 (PEER) Is Testing

Research question: Do CS faculty at mid-prestige R1 departments have the highest tenure rates, while those at elite departments face the same signal compression and slot scarcity seen in the Army and basketball settings?

Dimension	Army (CODA)	Basketball (SCOUT)	Academia (PEER)
Individual performance	OER evaluation score	Points per minute	Annual publications (OpenAlex)
Pool definition	Senior-rater cohort	Teammates	CS faculty cohort at same department × year
Peer quality measure	LOO pool mean OER	LOO mean teammate PPM	LOO mean publications per year (`poolq_loo_mean`)
Outcome	Promoted to Major	Drafted to NBA	Tenured (Assistant → Associate Professor)
Slot scarcity mechanism	Literal promotion quotas	Finite draft slots	Departmental FTE / budget lines

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3. What the Pipeline Built

Data source

Internet Archive (Wayback Machine) — CDX API to plan captures, then HTML downloads and custom parser (`html_parser.py`) to extract faculty name + rank from each archived page.

Coverage

- **Schools planned:** ~187 R1 CS departments in scope
- **Schools with usable scraped data:** ~60+ departments have parseable HTML in the current run
- **Time window:** approximately **2000–2023** (availability varies by school; earlier years are sparser)
- **Key limitation schools:** Auburn (robots.txt / Wayback exclusion), NC State (redirect shells), some schools with JS-heavy pages that CDX captured as near-empty HTML

Panel structure

Each row in the panel = **one person × one year** with: - Name, university, rank (assistant / associate / full / unknown / etc.) - Annual publications and cumulative publications (from OpenAlex, where matched) - Tenure outcome flags: `tenure_event`, `attrition`, `censored` - LOO peer pool metrics: `poolq_loo_mean`, `pool_rank_loo`, `pool_pctile_loo`, etc.

Panel size (as of May 2026 run): - `faculty_panel.jsonl`: ~1.7 GB / **~106,000 person-year rows** (includes all ranks, all confidence levels) - `faculty_panel_with_pools.jsonl`: 72 MB (enriched with pool quality metrics) - `faculty_panel_enriched.jsonl`: 55 MB (OpenAlex-matched subset)

OpenAlex linkage quality

Confidence	Meaning	Count (sample)
`HIGH`	Name + institution + year overlap	~32% of person-years
`MEDIUM`	Name + institution, year unclear	~15%
`MULTI`	Multiple plausible candidates — picked first	~19%
`LOW`	Name match only	~2%
`NONE`	No OA result	~32%

> ***Implication for VECTOR:*** *Publication data is missing or uncertain for roughly half the panel. This is a real limitation to acknowledge — the inverted-U is estimated on the subset with OA data. The advisor (Alex Gates, Apr 2026) was aware and directed moving forward with the existing data rather than waiting for bulk OA access.*

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4. The Inverted-U Finding (Stage 9 — Preliminary)

What was done

Faculty-years are **binned into 18 equal-width bins** of `poolq_loo_mean` (leave-one-out mean annual publications of same-dept assistant peers). For each bin, **tenure rate** = `n_tenure / n_resolved` (resolved = tenure + attrition; right-censored cases excluded from denominator).

Artifact

- **Plot:** `tenure_pipeline/stage9_inverted_u.png`
- **Data:** `tenure_pipeline/stage9_binned_table.csv`

Binned results (abbreviated)

Bin	LOO median pubs/yr	Tenure rate	N resolved
1 (lowest)	0.27	0.303	33
2	0.74	0.348	23
3	1.28	**0.488**	43
4–6	1.76–2.18	0.37–0.46	28–32
7	2.59	**0.537**	41
8–11	2.90–3.82	0.37–0.56	18–46
12–15	4.33–6.59	0.37–0.45	26–42
16	7.63	**0.667**	27
17	8.59	**0.697**	33
18 (highest)	12.72	**0.423** ← drop	26

What this shows

The pattern is **not a textbook symmetric inverted-U** — the full shape is: - **Low rates in the weakest pools** (bins 1–2: ~0.30–0.35) - **Rising through mid-range pools** (bins 3, 7, 9–11: ~0.49–0.56) - **High rates in strong pools** (bins 16–17: 0.67–0.70) - **A drop at the very highest quality pools** (bin 18: 0.42)

The **critical inverted-U feature** — a **peak followed by a decline at the top** — is present: tenure rates are **highest in the 85th–95th percentile** of pool quality, then **drop in the very elite tier** (bin 18 = top few percent of departments by peer publication rate). This matches the Army and basketball patterns structurally: the very best environments produce congestion / signal compression that lowers individual promotion probability.

Caveats to communicate to VECTOR

1. **High censoring:** ~50–60% of person-years are right-censored (still assistant near end of data window). Tenure rates are computed on resolved cases only; N per bin is small (~18–46 resolved cases).
2. **Preliminary / dirty-OK standard:** This is the "fast path first curve" per advisor guidance — not the final survival analysis. The Cox model (Cell 10 / 10.5) is the intended main specification.
3. **Pool quality measure is noisy:** Only faculty with HIGH or MEDIUM OpenAlex matches contribute to `poolq_loo_mean`. Departments with poor OA coverage have noisier pool quality estimates.
4. **Coverage is partial:** ~60 departments ≠ full R1 universe. Elite departments (top 10 CS programs) may be over- or under-represented depending on which schools have clean Wayback data.
5. **No controls yet:** The binned plot is **unconditional** — no cohort fixed effects, no school prestige controls, no individual ability controls. The shape could shift with proper modeling.

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5. What Is Wired But Not Yet Finalized

Element	Status
Cox survival model (Cell 10)	Code complete; time-varying covariates (`tb_ratio_fwd_snr`, pool metrics) specified in `COLUMN_CONFIG`; z-scored frame in Cell 10.5
Inverted-U test columns (`z_pool_minus_mean_snr_fwd`, `*_sq`, `star_pool_interaction`)	Created in Cell 10.5 on Cox-ready frame (not on raw snapshots — this is expected behavior, not a bug)
Formal model output	Not yet run / reported
School-level prestige covariate	Not yet merged into panel (planned: NRC rankings or USNews tier)
Attrition as competing risk	Flag exists in panel; competing-risks model not yet estimated

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6. How to Frame PEER in the Manuscript

What you CAN say (supported by current artifacts)

> *"In a preliminary analysis of CS faculty tenure at R1 universities, using archived faculty pages (Wayback Machine, 2000–2023) linked to OpenAlex publication records, we find a non-monotone relationship between peer pool quality and tenure rates consistent with the inverted-U predicted by the talent-pool congestion model. Tenure rates are lowest in the weakest departments, rise through mid-tier departments, and then decline in the most elite tier — mirroring the Army and basketball settings."*

What to hedge or defer

- Specific coefficient estimates (Cox model not finalized)
- Precise N of faculty or schools (coverage is still being QA'd)
- Claims about completeness (60+ schools, not the full 187-school R1 universe)
- Publication counts as "the" performance measure (OA linkage is partial)

Suggested status label for paper drafts

"Preliminary replication (Setting 3)." The inverted-U is visible in unconditional binned rates; formal survival analysis with pool-quality terms is the next step. Coverage limitations are acknowledged; full results are forthcoming pending pipeline completion and OpenAlex bulk access (pending UVA CDH provisioning).

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7. Key Files for VECTOR Reference

File	What it contains
`tenure_pipeline/stage9_inverted_u.png`	The inverted-U figure (pool quality bins vs tenure rate)
`tenure_pipeline/stage9_binned_table.csv`	Binned tenure rates with CIs (18 bins)
`tenure_pipeline/faculty_panel_with_pools.jsonl`	Full enriched panel (72 MB; all columns)
`tenure_pipeline/faculty_panel_advisor.csv`	Flat CSV version (columns reordered for readability)
`tenure/documents/PANEL_CSV_GLOSSARY.md`	Column-by-column definitions for the CSV
`tenure/documents/TENURE_DATA_GAMEPLAN.md`	Strategic contract: what PEER is trying to accomplish and why
`tenure/540_tenure_pipeline.ipynb`	Main pipeline notebook (Cells 0–10.5)

Document created 2026-06-03. Contact: Charles Levine / PEER agent.